NEURAL SPACETIME SEARCH WITH GRTECLYN: A PRACTICAL BASELINE FOR DYNAMICAL METRIC DISCOVERY

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We outline a practical baseline for turning **GRTeclyn** into a closed-loop GPU search engine for exotic spacetime metrics. A Python scheduler proposes compact initial-data parameters, a C++/AMReX executable evolves each candidate, and scoring diagnostics promote only geometries that survive, show operational faster-than-light (FTL) benefit, and avoid coordinate artifacts. Matter is handled either by algebraic reconstruction from the Einstein constraints at $t = 0$ or by a matter-first elliptic loop in which **GRTresna** solves the Hamiltonian and momentum constraints before **GRTeclyn** evolves the data.

I. INTRODUCTION

Most proposed interstellar metrics are hand-designed static ansätze. That approach is analytically clean but strongly biased toward highly symmetric geometries.

Here we treat metric discovery as black-box optimization over dynamical 3D initial data, searching for novel faster-than-light spacetime geometries—encompassing FTL transport, FTL communication, portal-like shortcuts, and wormhole throats—rather than restricting attention to any single warp ansatz. The framework couples a Python optimizer to the GPU AMR evolution engine in **GRTeclyn**, with explicit checks for matter consistency, stability, and fake-FTL gauge effects. A successful candidate must survive dynamical evolution, show an operational shortcut relative to a flat-spacetime control, keep tidal forces bounded, and minimize violations of the standard energy conditions.

Blind random 3D initial data almost always violates constraints, collapses, or relaxes to flat space. Optimization supplies the steering: maximize FTL benefit while penalizing constraint error, negative energy, tidal force, and instability.

This fits a broader shift toward automated theory exploration [1–4]. For warp geometries, WARP FACTORY and WARPAX emphasize observer-robust energy-condition evaluation [5, 6]; we extend that philosophy from analysis of prescribed metrics to search over evolved initial data, while keeping numerical convergence and constraint satisfaction central [7, 8].

Throughout this paper, "FTL" means effective faster-than-light propagation through a curved spacetime geometry compared with flat-spacetime light travel. It does not imply local superluminal motion relative to the local light cone.

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II. INITIAL DATA

The optimizer proposes the spatial metric γ_{ij} and extrinsic curvature K_{ij}; the Hamiltonian and momentum constraints then determine the compatible energy density ρ and momentum density j_i:

$$\rho = \frac{R + K^2 - K_{ij}K^{ij}}{16\pi}, \qquad j_i = \frac{1}{8\pi} D_j \left(K_i^j - \delta_i^j K \right). \tag{1}$$

Regions with $\rho < 0$ are recorded as exotic-matter cost and fed back into the objective. The implemented matter-first path using an elliptic constraint solver is described in Sec. V.

III. CANDIDATE PARAMETERIZATION

The optimizer emits a compact vector θ; a decoder maps it to smooth AMReX initial fields.

A. Level 1: Radial Recipes

The initial fields at $t = 0$ are built from spherically symmetric radial profiles:

$$q(r) = q_\infty + \sum_n a_n B_n(r), \tag{2}$$

where q denotes χ, α, β^i, or K, and B_n are smooth radial basis functions.

B. Level 2: Angular Recipes (implemented)

Angular modes add non-spherical structure:

$$q(r, \theta, \varphi) = q_\infty + \sum_{n\ell m} a_{n\ell m} B_n(r) Y_{\ell m}(\theta, \varphi). \tag{3}$$

This is implemented for gauge fields (α, β^i), where angular modes carry no constraint cost; geometric angular modes require the future 3D constraint solver.

C. Search space

The three FTL shortcut channels (shift-driven warp, $\chi > 1$ compression, areal-radius throat pinching) are independent. The default search vector is nine-dimensional: four χ coefficients, a shared basis width, two lapse coefficients, and two shift coefficients—all dimensionless amplitudes for the Gaussian basis expansion. This spans all three mechanisms while remaining tractable for CMA-ES. Higher-level parameterizations (neural decoders, LLM meta-agents) are deferred to Sec. VIII.

IV. THE SEARCH LOOP

The pipeline keeps Python orchestration separate from precompiled GPU evolution. The Python controller writes `params.txt` and dispatches Slurm tasks; GPU nodes run the same `GRTeclyn` binary for every candidate, avoiding recompilation inside the search loop.

Each optimizer evaluation follows:

1. propose basis coefficients θ;
2. derive constrained fields, write `params.txt`, and reject bad pre-flight residuals;
3. evolve on GPU and write constraint, collapse, matter, and plotfile diagnostics;
4. parse diagnostics into $S(\theta)$ and update CMA-ES after each generation.

The optimizer sees only the scalar black-box objective $f(\theta) = -S(\theta)$, making the score design the central physics interface.

Tier 1 uses short low-resolution single-GPU runs with early stopping; Tier 2 promotes survivors to multi-GPU resolution ladders for stability and convergence.

The current pipeline uses CMA-ES, not reinforcement learning: it samples continuous coefficients, ranks them by $-S(\theta)$, and updates a Gaussian search distribution. CMA-ES cannot add mouths, throats, or new mode families; RL becomes relevant only for sequential decisions such as choosing topology, modes, or mid-run gauge actions. A future hierarchical search could let a discrete controller choose topology and active modes, then hand the resulting skeleton to CMA-ES/MAP-Elites for continuous refinement, echoing discrete/continuous decompositions in string-landscape search [3].

V. THE INVERTED LOOP: MATTER-FIRST INITIAL DATA VIA GRTRESNA

The recipes of Sec. IV share a structural weakness that the results expose repeatedly: they manipulate the *geometry* and tolerate the constraint error it implies. This section describes an alternative path—an in-the-loop elliptic constraint solver (`GRTresna`)—that inverts the map, the physical motivation for it, the matter basis it searches over, and the current clean-search findings.

A. Geometry-first versus matter-first

The Level-1/2 recipes are *geometry-first*: the optimizer proposes χ, α, β^i, K directly and the constraints are read *backwards* (Eq. 40–41) to report whatever energy density ρ and momentum density j_i that geometry would require. This is cheap— it is algebraic and evaluated once at $t = 0$—but it has two structural limits. First, a non-spherical *geometric* deformation does not actually solve the Hamiltonian constraint; only the gauge sector (α, β^i) is free of constraint cost, which is precisely why the directional winner of Sec. VII lives entirely in the shift. Second, the constrained radial recipe uses $K = \text{const}$, $\Pi = 0$, which forces the momentum density to vanish, $j_i = 0$: the recipe *cannot represent moving matter*, and a translating warp needs exactly that momentum source (the "motor" the dipole winner was missing, Sec. VII).

`GRTresna` is a Chombo/AMR multigrid solver that inverts the map. It is *matter-first*: the optimizer proposes a *matter* configuration— a scalar field (ϕ, Π)—and the solver *solves* the full elliptic Hamiltonian and momentum constraints in 3D for a constraint-satisfying geometry. In the conformal transverse-traceless decomposition $\gamma_{ij} = \psi^4 \delta_{ij}$, $K_{ij} = \psi^{-2} \bar{A}_{ij} + \frac{1}{3}\gamma_{ij} K$, with the scalar sources

$$\rho = \tfrac{1}{2}\psi^{-4}|\tilde{\nabla}\phi|^2 + \tfrac{1}{2}\Pi^2 + V(\phi), \qquad S_i = -\,\Pi\,\partial_i\phi, \quad (4)$$

the solver (in its CTTKHybrid variant) fixes the trace by $K^2 = 24\pi G\rho$ so the matter cancels in the Hamiltonian, then solves

$$\tilde{\nabla}^2\psi = -\tfrac{1}{8}\,\bar{A}_{ij}\bar{A}^{ij}\,\psi^{-7}, \quad (5)$$

$$\bar{A}_{ij} = (\mathbb{L}V)_{ij}, \quad \tilde{\nabla}^2 V_i + \tfrac{1}{3}\partial_i(\tilde{\nabla}\cdot V) = \psi^6\left(\tfrac{2}{3}\partial_i K + 8\pi G\, S_i\right), \quad (6)$$

for the conformal factor ψ and a vector potential V_i whose longitudinal derivative $(\mathbb{L}V)_{ij}$ is the transverse-traceless $\bar{A}_{ij}$. We *modify the matter*; the solver returns the geometry. The result is data that satisfies $\|H\|_{L^2}, \|M_i\|_{L^2} \sim 10^{-3}$–$10^{-4}$ in full 3D—roughly $100\times$ tighter than the recipe—so the dynamics that are scored are physical rather than constraint-relaxation transients.

B. The GRTresna ↔ GRTeclyn bridge

Each optimizer evaluation now contains a nested elliptic solve. A Python orchestrator writes the `GRTresna` parameter file, launches the MPI multigrid solve, and converts its Chombo HDF5 output into a compact binary (`.gridinit`) sampled on the target Cartesian grid; `GRTeclyn` ingests it through a new `ExternalGridInitialData` loader (`recipe_initial_data_file`) and evolves it on the GPU. No GRTeclyn evolution code is special-cased: it computes the same momentum density $j_i = -\Pi\,\partial_i\phi$ and evolves the full CCZ4 + Klein–Gordon system, so the solved $\bar{A}_{ij}$ plus the Γ-driver shift develop the frame-dragging shift dynamically. The heavy intermediate HDF5 is deleted after conversion, keeping a long conveyor search disk-bounded.

C. The scalar-lump matter basis

The matter configuration is a superposition of K localized scalar-field *lumps*. Lump k is a Gaussian-enveloped, optionally anisotropic profile

$$\phi_k(\vec{x}) = A_k\, a_{m_k}(\vec{x})\, \exp\left[-\frac{|\vec{x}-\vec{c}_k|^2}{2w_k^2}\right], \tag{7}$$

where a_m is a Cartesian azimuthal factor ($a_0 = 1$; $a_1 = \delta x/w$ dipole; $a_2 = (\delta x^2 - \delta y^2)/w^2$ quadrupole) chosen to avoid the z-axis coordinate singularity of $\cos m\varphi$. The conjugate momentum is the convective derivative of a pattern boosted with velocity $\vec{v}_k$ and rigidly rotated about z at rate ω_k,

$$\Pi_k = -\,\vec{v}_k\cdot\tilde{\nabla}\phi_k \;-\; \omega_k\,\partial_\varphi\phi_k, \tag{8}$$

so that via Eq. (4) the configuration carries net linear momentum $P_i \sim v_i$ and (for $m \geq 1$) net angular momentum $L_z \sim \omega$. Each lump therefore exposes eleven search dimensions—amplitude A_k, width w_k, center $\vec{c}_k$ (3), boost $\vec{v}_k$ (3), rotation ω_k, integer azimuthal mode m_k, and an exotic flag (Sec. V G)—and the default $K = 5$ basis spans a 55-dimensional matter search. The production search keeps the clouds compact (widths 1.5–4.5 and centers within a small central box) so the optimizer learns localized matter configurations rather than painting the whole frame. A superposition of lumps paints an essentially arbitrary energy/momentum distribution that the elliptic solve maps to a rich family of constraint-satisfying geometries (χ wells plus momentum-driven $\bar{A}_{ij}$); two counter-moving lumps, for example, give a strong momentum/shear field with no bulk translation. Black holes are disabled in the search configuration so the optimizer explores pure moving/rotating scalar clouds.

D. Parameter reduction by a structured ring ansatz

The fully free five-lump basis is expressive but expensive for CMA-ES. With $K = 5$ it exposes 55 continuous/categorical coordinates, while each evaluation requires a full MPI elliptic solve and, for survivors, a GPU evolution. In this regime CMA-ES can learn broad facts—for example that non-convergent exotic configurations are bad—but it wastes many samples distinguishing arbitrary translations, rotations, and permutations of essentially the same matter pattern. The optimization problem is therefore better posed in terms of coherent *matter currents* than independent blobs.

We introduced a reduced `ring` parameterization that searches a small set of global template variables and then deterministically expands them back into the same lump representation consumed by `GRTresna`. For K generated lumps, their angular positions are

$$\vartheta_k = \Phi + \frac{2\pi k}{K}, \qquad \vec{c}_k = R(\cos\vartheta_k, \sin\vartheta_k, 0) + z_0 \sin(\vartheta_k - \Phi)\,\hat{z}, \tag{9}$$

and their velocity is decomposed into radial, tangential, and vertical pieces,

$$\vec{v}_k = v_r\,\hat{r}_k + v_\varphi\,\hat{\varphi}_k + v_z \cos(\vartheta_k - \Phi)\,\hat{z}. \tag{10}$$

The amplitude and width are modulated by low Fourier modes,

$$A_k = A_0[1 + d\cos\vartheta_k + q\cos(2\vartheta_k)]_+, \qquad w_k = w_0\left[1 + \tfrac{1}{4}q\sin(2\vartheta_k)\right]_+ \tag{11}$$

with clipping to the same safe bounds as the free-lump search. A separate exotic fraction and exotic phase choose which sector of the ring is phantom: the $N_{\text{exotic}} \simeq f_{\text{ex}}K$ lumps with largest $\cos(\vartheta_k - \Phi_{\text{ex}})$ are assigned the exotic sign. The remaining searched variables are the shared internal rotation ω and azimuthal mode m.

Thus the default matter-first search now uses fourteen parameters:

$$\{A_0, w_0, R, z_0, \Phi, v_\varphi, v_r, v_z, \omega, f_{\text{ex}}, \Phi_{\text{ex}}, d, q, m\},$$

instead of the $11K = 55$ free-lump coordinates. The reduction is not a loss of the physical ingredients we need: the ansatz still contains compact scalar matter, angular momentum, counterflow/shear, localized exotic support, and dipole/quadrupole asymmetry. It merely ties together degrees of freedom that are related by an approximate rotating-ring symmetry. In physical terms, the optimizer searches for a rotating or shearing scalar-current loop—possibly with an exotic sector on one side—rather than a cloud of unrelated matter blobs. This keeps the downstream physics unchanged (the expanded `cfg.lumps` are passed through the same elliptic solve and GPU evolution) while giving CMA-ES a far lower-dimensional landscape to learn.

E. From a planar ring to a full-sphere discovery ansatz

The ring of Eq. (9) is a strong, deliberately chosen prior: its centres lie on a single equatorial circle (plus a small z_0 wobble), so it is efficient *refinement* for one physically motivated family—a rotating matter-current loop, whose tangential flow sources the angular momentum and frame-dragging "motor" that the dipole winner of Sec. VII was missing. But because we do not know *a priori* what an FTL configuration looks like, a prior that cannot leave one plane is a poor *discovery* tool: it structurally excludes matter or currents distributed elsewhere on the 2-sphere (axial dipoles, shells, obliquely oriented loops).

We therefore add a complementary `shell` ansatz that retains the same low-dimensional, current-based philosophy but covers the full sphere. The K lumps are placed on a deterministic Fibonacci lattice and rigidly rotated to a searched orientation axis $\hat{a}(\theta_a, \varphi_a)$, and the matter current is decomposed into the two families a sphere supports—a *toroidal* component circulating about $\hat{a}$ (net angular momentum $L_{\hat{a}}$) and a *poloidal* component flowing over the poles:

$$\vec{v}_k = v_r\,\hat{r}_k + v_t\,\frac{\hat{a}\times\hat{r}_k}{|\hat{a}\times\hat{r}_k|} + v_p\,(\hat{t}_k\times\hat{r}_k), \tag{12}$$

with amplitude and width modulated along the axis by low Legendre orders P_1, P_2 and a longitudinal exotic wedge. This adds only two coordinates over the ring (16 versus 14) while making the planar ring a strict thin-equatorial-band limit ($v_p\to 0$, $\hat{a}=\hat{z}$). The intended workflow is hierarchical and matches the discrete/continuous decomposition of Sec. IV: the broad `shell`/`free` bases (and the MAP-Elites driver) *discover* candidate families across the sphere, and the reduced `ring` then *refines* a chosen family cheaply. Crucially the expensive part is unchanged—every ansatz expands to the same `cfg.lumps` consumed by the identical elliptic solve and GPU evolution—so broadening coverage costs optimizer dimensions, not solver complexity.

F. What the elliptic solve adds to the search

Relative to the geometry-first recipes, the matter-first path contributes three genuinely new ingredients. (i) *Momentum density.* The recipe has $S_i = 0$; the momentum constraint Eq. (6) solves for $S_i = -\Pi\,\partial_i\phi \neq 0$, the source of matter-momentum-driven frame dragging—a distinct, physically grounded operational-FTL axis that was simply absent from the search space before. (ii) *Constraint-correct non-spherical structure.* Because the Hamiltonian is solved in 3D rather than along a 1D ray, directional structure in χ and $\bar{A}_{ij}$ no longer reintroduces off-axis constraint error. (iii) *Trustworthy dynamics.* The $\sim 100\times$ lower initial constraint error suppresses the junk gravitational radiation that otherwise contaminates the stability and growth-rate metrics.

G. Exotic matter and the maximal-slicing solver

A persistent FTL channel generically needs *exotic* matter ($\rho < 0$, an NEC violation), but the CTTK family is structurally positive-energy: the trace fix $K = \mathrm{sgn}\sqrt{24\pi G\rho + \ldots}$ has a negative argument wherever $\rho < 0$, so any non-trivial exotic source yields $K \in \mathbb{C}$ and the solve returns `NaN`. We added (a) a per-lump phantom flag—an independent-field stress tensor in which an exotic lump's kinetic and momentum contributions enter Eq. (4) with a flipped sign, so a configuration can mix normal and exotic lumps—and (b) a *maximal-slicing* ($K = 0$) path that, instead of absorbing the matter into K, sources it elliptically:

$$\tilde{\nabla}^2\psi = -2\pi G\rho\,\psi^5 - \tfrac{1}{8}\bar{A}_{ij}\bar{A}^{ij}\,\psi^{-7}, \tag{13}$$

solved by an under-relaxed Newton iteration with a $\psi > 0$ floor for the resulting indefinite operator. To make this indefinite solve survive adaptive mesh refinement we add three multilevel safeguards: (i) the matter contribution to the Newton Jacobian is capped and ψ is floored finite and positive, so a single bad cell cannot poison the multigrid V-cycle; (ii) coefficient coarsening switches from harmonic to *arithmetic* averaging, which stays well defined where the sign-changing a-coefficient straddles zero across a coarse–fine face; and (iii) refinement tagging keys on $|\rho|$ so negative-energy regions are refined rather than cancelled by neighbouring positive matter. The result is correct and robust *under full AMR*: canonical matter reproduces the standard path, a single isolated exotic lump converges to finite constraint-satisfying data where the old solver gave only `NaN`, and—most importantly—the production failure mode (three overlapping lumps, mixed canonical/exotic, angular mode $m = 1$, `max_level` $= 3$) now converges cleanly: across the canonical, isolated-exotic, and mixed `max_level`$= 3$ smoke solves the Hamiltonian/momentum residuals are 0.0056%/0.025%, 0.099%/0.027%, and 0.058%/0.029% respectively, with no `NaN`. A sharp residual cliff at high exotic amplitude marks the Lichnerowicz/York *existence boundary*: beyond a critical negative-energy density no asymptotically-flat constraint-satisfying solution exists, a physical rather than numerical limit.

H. Closed-loop controls and clean-search result

The matter-first loop is now the production path rather than a deferred item. Each candidate follows

$$\theta_{\text{matter}} \to \texttt{GRTresna} \to \texttt{.gridinit} \to \texttt{GRTeclyn} \to S(\theta), \tag{14}$$

where the optimizer sees eleven variables per lump and a five-lump basis by default in the fully free mode, the fourteen ring-template variables of Sec. V D in the reduced refinement mode, or the sixteen full-sphere `shell` variables of Sec. V E in the discovery mode. Exotic candidates are routed onto the hardened $K = 0$ path, while purely canonical candidates may retain the trace-absorption path. The search objective is FTL-first: evolved operational FTL and the local cone-tilting precursor are ranked ahead of health terms, while random and forced-exotic injections prevent CMA-ES from collapsing too early to one matter pattern.

The essential numerical control is a *GRTresna quality gate*. If the elliptic solve returns missing, non-finite, or large Hamiltonian/momentum residuals, the candidate is rejected before the GPU evolution. The rejection is not a flat score: its fitness penalty grows with the Ham/Mom residuals, so CMA-ES can learn that a marginally non-converged solve is less bad than a `NaN` or `Ham`=100% candidate. This removed the earlier failure mode in which nonconverged initial data could still produce a misleading high FTL score after `GRTeclyn` crashed at the first step.

A clean gated 8-GPU search (`runs/grtresna_search/optimize_20260602T181607Z`) produced 88 proposed candidates before being stopped for inspection: 54 completed GPU evolutions and 26 bad `GRTresna` solves were rejected before evolution. The best candidate, `eval_000063`, used three exotic lumps out of five and converged in `GRTresna` with Ham/Mom residuals 0.1676%/0.0048%. It showed no initial/static shortcut ($F_{\rm shortcut} = 0$), but after evolution the operational probe found

$$F^{\rm ev}_{\rm op} = 0.0461, \qquad \max c_{\rm coord} = 1.0888, \qquad t_{\rm min} = 15.024 < t_{\rm flat} = 15.75.$$

Thus the current matter-first search has found a compact exotic-matter configuration with a genuine evolved coordinate-superluminal channel. This is a promising FTL precursor, not yet a claimed physical warp drive: the candidate still requires deterministic replay, longer evolution, and resolution convergence before promotion.

VI. SCORE COMPONENTS AND WHY THEY MATTER

The central challenge is to discover spacetime metrics that exhibit specific physical qualities: effective FTL shortcuts, dynamical stability, bounded tidal forces, and minimal exotic matter. The score function must reward these qualities while ensuring that trivial Minkowski spacetime—which is perfectly stable and constraint-clean—does not dominate the search. This section defines every component of the scoring objective in one place.

A. Bounded reward and total score

Each raw diagnostic value v_k (such as a constraint norm, a growth rate, or an exotic-matter integral) is mapped to a bounded reward $s_k \in [0,1]$ via

$$s_k = r(v_k;\, \sigma_k) = \frac{1}{1 + v_k/\sigma_k}, \tag{15}$$

where $v_k \geq 0$ is the diagnostic value (lower is better for penalties) and σ_k is a characteristic scale that controls sensitivity—when $v_k = \sigma_k$, the reward is exactly $\frac{1}{2}$. The total score is the weighted sum

$$S(\theta) = \sum_k w_k\, s_k(\theta). \tag{16}$$

This mapping keeps every component in $[0,1]$ regardless of physical units, while the weights w_k (Table I) set the relative importance.

B. FTL shortcut metrics

Three independent shortcut indicators are evaluated on the x-axis from the $t = 0$ profiles.

a. Warp null-ray shortcut. For conformally flat slices, an axial null ray has coordinate speed $dx/dt = \alpha(x)\sqrt{\chi(x)} - \beta^x(x)$, giving travel time

$$T_{\rm curved} = \int_{-L}^{L} \frac{dx}{\alpha\sqrt{\chi} - \beta^x}, \qquad T_{\rm flat} = 2L, \tag{17}$$

and reward

$$F_{\rm null} = \max\left(0,\, \frac{T_{\rm flat} - T_{\rm curved}}{T_{\rm flat}}\right). \tag{18}$$

Paths below a coordinate-speed floor are rejected.

b. Portal proper-distance shortcut. The portal-like compression reward is

$$D_{\rm proper} = \int_{-L}^{L} \chi^{-1/2}(r)\, dr,$$
$$F_{\rm portal} = \max\left(0,\, \frac{2L - D_{\rm proper}}{2L}\right). \tag{19}$$

c. Wormhole throat pinch. The throat reward compares an interior areal-radius minimum to the asymptotic radius:

$$F_{\rm throat} = \max\left(0,\, 1 - \frac{R_{\rm min}}{R_{\rm asymp}}\right). \tag{20}$$

d. Anti-flatness gate and combined FTL objective. The anti-flat gate ensures only non-trivial geometries score:

$$s_{\rm nonflat} = \min\left(1,\, \frac{\max|\chi - 1| + \max|\beta^x|}{\tau}\right), \tag{21}$$

with scale $\tau \sim 0.08$, giving

$$F_{\rm FTL} = \max(F_{\rm null}, F_{\rm portal}, F_{\rm throat}) \cdot s_{\rm nonflat}. \tag{22}$$

e. Natário–Alcubierre expansion asymmetry. For a warp-like contraction-ahead/expansion-behind pattern:

$$F_{\rm asymmetry} = \max\left(0,\, \frac{\int_{x>0} -\theta(\mathbf{x})\, d^3x + \int_{x<0} \theta(\mathbf{x})\, d^3x}{\int |\theta(\mathbf{x})|\, d^3x + \epsilon_\theta}\right). \tag{23}$$

The implemented 1D evaluator reduces these integrals to trapezoidal sums along the travel axis.

f. Logarithmic FTL amplification. Weak shortcuts are log-amplified so even a few-percent shortcut becomes optimizer-visible:

$$F_{\log} = \frac{\ln(1+\eta F_{\rm FTL})}{\ln(1+\eta)}, \tag{24}$$

with $\eta \approx 100$. The optimizer uses $F_{\log}$ rather than raw $F_{\rm FTL}$.

g. Operational probe FTL. A major risk is gauge-driven fake FTL. High-scoring candidates must therefore pass an operational arrival-time test against a flat control on the same grid, with no trapped-surface crossing and bounded constraints:

$$F_{\rm op} = \frac{t_B^{\rm flat} - t_B^{\rm curved}}{t_B^{\rm flat}}, \tag{25}$$

accepted only when constraints remain bounded and no trapped surface is crossed.

C. Health and stability metrics

a. Co-moving stability. Eulerian stability penalizes moving bubbles, so stability is estimated in a co-moving frame. The average axial shift is

$$\langle \beta^x \rangle = \frac{1}{V_{\rm bubble}} \int_{\rm bubble} \beta^x(\mathbf{x})\, d^3x, \tag{26}$$

and the co-moving budget is

$$\begin{aligned} \Delta_{\rm comoving} &= \frac{\max_t |\partial\chi/\partial t|}{\max(|\langle\beta^x\rangle|,\, 10^{-3})}, \\ s_{\rm comoving} &= \frac{1}{1+\Delta_{\rm comoving}}. \end{aligned} \tag{27}$$

For stationary throats or portals with negligible shift, the scorer falls back to Eulerian stability.

b. Dynamical stability and geometry growth. Short runs can survive while already focusing toward collapse. We therefore score geometry growth explicitly. For each diagnostic q (from $\alpha_{\min}$, $\chi_{\min}$, $|K|_{\max}$, r_H, $R_{\rm areal}$), fractional violation terms are

$$\begin{aligned} \Delta q_{\rm grow} &\equiv \frac{\max_t q(t) - q(0)}{\max\big(|q(0)|,\, q_{\rm floor}\big)}, \\ \Delta q_{\rm drop} &\equiv \frac{q(0) - \min_t q(t)}{\max\big(|q(0)|,\, q_{\rm floor}\big)}, \end{aligned} \tag{28}$$

with terms clipped to ≥ 0. The implemented indicators are:

$$\Delta_K \equiv \Delta\big(|K|_{\max}\big)_{\rm grow} \quad \text{with } q_{\rm floor} = 10^{-1}, \tag{29}$$

$$\Delta_\alpha \equiv \Delta\big(\alpha_{\min}\big)_{\rm drop} \quad \text{with } q_{\rm floor} = 10^{-6}, \tag{30}$$

$$\Delta_\chi \equiv \Delta\big(\chi_{\min}\big)_{\rm drop} \quad \text{with } q_{\rm floor} = 10^{-8}, \tag{31}$$

$$\Delta_{r_H} \equiv \Delta\big(r_H\big)_{\rm grow} \quad \text{with } q_{\rm floor} = 1, \tag{32}$$

$$\Delta_{R_{\rm areal}} \equiv \max_t \frac{\big|R_{\rm areal}(t) - R_{\rm areal}(0)\big|}{\max\big(|R_{\rm areal}(0)|,\, 10^{-8}\big)}. \tag{33}$$

The geometry-change budget $\Delta_{\rm geom} = \Delta_K + \Delta_\alpha + \Delta_\chi + \Delta_{r_H} + \Delta_{R_{\rm areal}}$ gives the stability reward

$$s_{\rm stability} = r(\Delta_{\rm geom};\, 1) = \frac{1}{1+\Delta_{\rm geom}}. \tag{34}$$

Higher is better; $s_{\rm stability} < 0.25$ is treated as a promotion failure. A good equilibrium should approach $s_{\rm stability} \gtrsim 0.8$.

c. Constraint growth rate. Late-time diagnostics are fit to $q(t) \sim q_0 e^{\lambda t}$ and the worst growth is penalized:

$$\begin{aligned} s_{\rm growth} &= r(\max(0, \lambda_{\rm eff});\, \sigma_\lambda)\,, \\ \lambda_{\rm eff} &= \max\big(\lambda_{\|H\|},\, \lambda_{|K|_{\max}},\, \lambda_{1/\chi_{\min}}\big), \end{aligned} \tag{35}$$

with $\sigma_\lambda = 0.5$.

D. Risk penalties

a. ANEC line proxy. The required energy is integrated along the travel axis:

$$\begin{aligned} \mathcal{A}_{\rm line} &= \int_{-L}^{L} \rho_{\rm req}(x)\, dx, \\ s_{\rm anec} &= r(\max(0, -\mathcal{A}_{\rm line});\, \sigma_{\mathcal{A}})\,, \end{aligned} \tag{36}$$

with $\rho_{\rm req} = (R + \frac{2}{3}K^2)/16\pi$ and $\sigma_{\mathcal{A}} = 0.1$. The full averaged null energy condition target is

$$\mathcal{A} = \int_\gamma T_{\mu\nu}\, k^\mu k^\nu\, d\lambda, \tag{37}$$

which requires the evolved probe and a local tetrad.

b. Tidal proxy. $T_{\rm tidal} = \max_x |R(x)| + \max_x |\partial_x^2 \alpha(x)|$, combined with a $t = 0$ trapped-surface flag until full 4-curvature diagnostics are available.

c. Other risk terms. The score also includes $s_{\rm horizon}$ (penalizing trapped-surface formation), $s_{\rm EC}$ (penalizing energy-condition violations from the evolved stress-energy), and $s^{\rm eff}_{\rm exotic}$ (a Hawking–Ellis/Type-I hybrid margin from the evolved 4-metric following WARPAX [6], computed with fourth-order finite differences [7]).

E. The non-triviality gate

Flat Minkowski spacetime scores perfectly on every health metric. To prevent it from dominating, health rewards are multiplied by a non-triviality gate:

$$g_{\rm nt} = \min\Big(1,\ \max\big(s_{\rm nonflat},\, F_{\rm asymmetry},\, s_{\rm nontrivial},\, F_{\rm curv},\, F_{\rm op},\, F_{\log},\, s^{\rm eff}_{\rm exotic}\big)\Big) \in [0,1]. \tag{38}$$

The health rewards $\mathcal{H}$ are gated, while FTL/exoticity terms remain ungated:

$$S(\theta) = g_{\rm nt}(\theta) \sum_{k\in\mathcal{H}} w_k s_k(\theta) \ + \sum_{k\notin\mathcal{H}} w_k s_k(\theta). \tag{39}$$

With this gate, flat Minkowski drops to $S = 0$ while structured reference geometries keep their original scores. Unavailable diagnostics contribute zero, so legacy episodes re-score consistently.

F. Constraint-satisfying recipe

For the conformally flat recipe ($\tilde{\gamma}_{ij} = \delta_{ij}$, $A_{ij} = 0$), the constraints reduce to

$$R[\chi] + \tfrac{2}{3}K^2 = 16\pi\rho, \tag{40}$$

$$-\tfrac{2}{3}D^i K = 8\pi j^i. \tag{41}$$

With $\Pi = 0$, the momentum constraint requires spatially constant K; the scalar field ϕ is then derived from χ to reduce Hamiltonian violation. The pipeline consists of `constrained_recipe.py` (derive ϕ from χ with $K = \text{const}$, $\Pi = 0$), `preflight.py` (reject large residuals or negative χ before GPU launch), and `RadialRecipeLevel.cpp` (write ρ_{req} and constraint diagnostics).

A test suite passes **74/74 assertions**: flat space remains exact, $K = \text{const}, \Pi = 0$ drives $\|M_i\|_{L^2} = 0$, pathological inputs are rejected, and constrained overrides usually reduce $\|H\|_{L^2}$. On 27 synthetic profiles, the batch harness finds 7 `accepted_phantom`, 19 `rejected_exotic_energy`, and 1 `rejected_negative_chi`.

The first symmetry-breaking extension adds axisymmetric deformations,

$$\chi(r,\theta) = \chi_0(r) + \sum_a A_a\, e^{-(r-r_a)^2/2w_a^2}\, P_{\ell_a}(\cos\theta), \tag{42}$$

with a ray check to ensure $\chi > 0$.

VII. RESULTS

A. Reference seed validation

The FTL scoring formulas were validated on analytical reference seeds at $L = 8$. Flat Minkowski scores $F_{\text{FTL}} = 0$ across all channels. The logarithmic amplification makes subluminal Alcubierre signals optimizer-visible ($F_{\log} = 0.43$–0.57 for $v = 0.3$–0.8), while Ellis–Bronnikov and Schwarzschild score through the throat and proper-distance channels ($F_{\text{FTL}} > 0.87$) even with $F_{\text{null}} = 0$. On the risk side, Alcubierre seeds have zero ANEC violation and zero tidal curvature at $t = 0$ (pure-shift warp), whereas Ellis–Bronnikov and Schwarzschild carry large axis exotic-energy integrals ($\mathcal{A}_{\text{line}} = -6.6$ and -2.4) and trapped-surface flags.

B. GPU smoke validation

Short $t = 2$ GPU runs at $N_{\text{full}} = 64$ validated the end-to-end pipeline. Three spherical candidates survived with controlled 3D constraint norms ($\|H\|_{L^2} \leq 0.042$), confirming that the 1D pre-flight is conservative. Stability scores ranged from $s_{\text{stability}} = 0.30$ (Ellis–Bronnikov) to 0.68 (bubble-wall profile). Extended $t = 5$ tests promoted Ellis–Bronnikov and the bubble-wall seed to the resolution ladder.

C. Non-spherical candidates

All seven ray-sane non-spherical candidates (dipole, quadrupole, mixed-mode, and random angular deformations) survived short GPU evolution with controlled constraints ($\|H\|_{L^2} \leq 0.014$, $s_{\text{stability}} \approx 0.44$–$0.51$). However, these χ-only angular deformations produce strong expansion asymmetry ($F_{\text{asym}} \approx 0.99$) but zero axial null shortcut ($F_{\log} = 0$). Angular χ deformations plateau on asymmetry alone; shift-enabled warps are needed to exceed flat space.

D. Stability and growth analysis

At $t = 2$, the upgraded FTL scoring resolves the flat-space attractor: Alcubierre $v = 0.5$ ($S = 10.08$) narrowly outranks flat Minkowski ($S = 10.00$) because $F_{\log}$ and s_{comoving} offset the Eulerian stability penalty, while Ellis–Bronnikov scores highest ($S = 10.84$) through its throat channel.

However, every short-run survivor still has positive exponential growth rate λ_{eff}, ranging from 0.29 (random seeds) to 1.02 (Ellis–Bronnikov), exposing the central stability gap: short-run survival does not imply long-term equilibrium. The first full-domain AMR CMA-ES campaign confirmed this—all top candidates had $s_{\text{stability}} < 0.09$, showing that the optimizer exploits the short stop time rather than finding genuinely stable geometries. This motivates the s_{growth} penalty and longer-run validation below.

E. Longer-run validation

Longer $t = 16$ runs (Table II) test the exotic scalar matter fix and evolved operational FTL probe beyond the smoke window. Three main implications emerge. First, the exotic field produces the expected matter NEC violations for Ellis–Bronnikov and mild non-spherical phantom candidates. Second, effective energy-condition evaluation of $T^{\text{eff}}_{\mu\nu} = G_{\mu\nu}/8\pi$ recovers the geometric exotic energy of shift-driven Alcubierre data even when scalar-matter NEC is zero. Third, evolved-frame FTL is essential: Alcubierre's initial shortcut decays, while Ellis–Bronnikov develops an evolved shortcut absent on the $t = 0$ slice. The gated objective removes flat Minkowski ($S = 0$) without changing structured candidates.

TABLE I. Compact score-weight summary. Components are inactive when their diagnostics are unavailable.

Group	Weights
FTL value	$5F_{\text{log}} + 2F_{\text{asymmetry}} + s_{\text{nonflat}} + 3F_{\text{op}}$
Health	$1.5s_{\text{survival}} + 2s_{\text{constraints}} + s_{\text{lapse}} + 2.5s_{\text{comoving}} + 0.5s_{\text{stability}}$
Risk penalties	$1.5s_{\text{horizon}} + 2s_{\text{EC}} + 2s_{\text{growth}} + 1.5s_{\text{anec}} + s_{\text{tidal}}$
Evolved geometry	$0.5F_{\text{curv}} + 1.5s^{\text{eff}}_{\text{exotic}}$

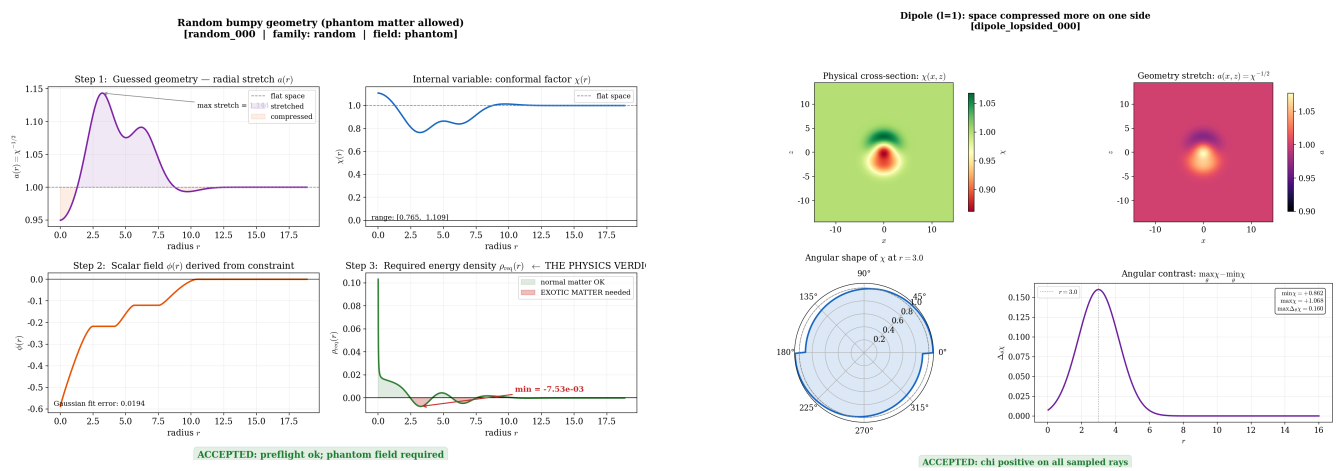

FIG. 1. **Metric-guesser validation examples.** Left: accepted phantom random candidate, showing guessed stretch $a(r) = \chi^{-1/2}$, stored $\chi(r)$, derived scalar $\phi(r)$, and required energy density $\rho_{\text{req}}(r)$; it is numerically constructible but accepted only in the phantom/exotic-matter class because $\rho_{\text{req}} < 0$ in places. Right: accepted non-spherical dipole candidate; the $\ell = 1$ mode makes the geometry lopsided while χ stays positive throughout the ray check.

F. Search optimization

Beyond CMA-ES, two complementary strategies were tested. A surrogate-assisted RBF kernel-ridge model pre-screens proposals, skipping 27% of GPU evaluations in one 8-GPU run while retaining the best score. MAP-Elites over FTL benefit and exoticity produces a diverse failure atlas (7 distinct elites from 40 evaluations on a 6×6 grid, scores $S = 9.9$–12.5) rather than a single optimum. Pareto extraction over $(F_{\text{FTL}}, s_{\text{anec}}, s_{\text{growth}}, s_{\text{tidal}})$ exposes trade-offs hidden by scalar weights.

G. The dipole metric: gauge-angular search winner

Because angular χ deformations cannot be made constraint-consistent by a 1D solve, we opened the gauge sector: α and β^i carry Legendre modes with $\ell \in \{1, 2\}$ and searchable radial centers/widths, giving a 21-parameter gauge-angular space. An 8-GPU campaign found evolved superluminal channels at $\int \rho_- \, dV \approx 0.13$–$0.17$, a factor 2–5 below spherical persistent-FTL candidates.

The three top-ranked candidates of the campaign and the long-run validation of the best are compared against the reference solutions by gated score S in Table III. The top candidate, `eval_000188`, was promoted and re-evolved to $t = 16$ at $N_{\text{full}} = 192$, `max_level=2` (Fig. 2). The validated geometry is a standing Alcubierre-like χ dipole, not a translating bubble: the evolved channel persists weakly ($\max c : 2.67 \to 1.32$, $F^{\text{ev}}_{\text{op}} = 5.6 \times 10^{-3}$, 51% of the path superluminal), constraints stay clean and decreasing ($\|H\|_{L^2} : 1.1 \times 10^{-3} \to 4.2 \times 10^{-4}$, $\|M\|^{\text{fin}}_{L^2} = 3.6 \times 10^{-5}$), and the required exotic matter stays small and concentrated in a single localized core at the compression lobe ($\int \rho_- \, dV = 0.13$, $\min \rho_{\text{req}} \approx -1.0 \times 10^{-3}$, ANEC line -6.6×10^{-3}; Fig. 2), though a near-trapped flag appears in the compressed lobe ($\min \theta_+ = -0.11$).

Even after honest long-time validation (Table III), the dipole metric achieves a gated total $S = 16.75$, outperforming both Alcubierre $v = 0.5$ ($S = 14.9$) and Ellis–Bronnikov ($S = 12.8$) evaluated on the same $t = 16$ window. Crucially, it accomplishes this with an exotic-matter cost of $\int \rho_- \, dV = 0.13$ on a *normal*-matter solve—substantially lower than spherical persistent-FTL candidates, and without the phantom matter the reference wormhole seeds require—while maintaining a superluminal path fraction of 0.51 in the evolved frame. This demonstrates that the automated search can discover geometries that outperform hand-designed warp metrics on the combined objective of FTL benefit, stability, and energy-condition cost.

a. Where the exotic matter actually lives. The integrated cost $\int \rho_- \, dV$ is a single scalar; to see *where* energy-

TABLE II. Longer-run validation ($t = 16$, $N_{\rm full} = 64$, $L = 8$, H100) with the gated objective (Eq. 39). $F^0_{\rm op}/F^{\rm ev}_{\rm op}$ are the operational FTL benefits on the $t = 0$ slice and the evolved plotfile; $\mathrm{NEC}^{\rm mat}$ is the minimum matter null-energy margin over the run (negative = violation); $g_{\rm nt}$ is the non-triviality gate. S is the gated total: flat Minkowski collapses to 0 while every non-trivial geometry is unchanged. All reference runs use constrained phantom initial data.

Example	survival	$s_{\rm stab}$	$F^0_{\rm op}$	$F^{\rm ev}_{\rm op}$	$\mathrm{NEC}^{\rm mat}$	$g_{\rm nt}$	S
Flat Minkowski	1.00	1.00	0	—	0	0.0	**0.0**
Alcubierre $v = 0.5$	1.00	0.02	0.096	0.0005	0	1.0	**14.9**
Ellis–Bronnikov (N=16)	0.80	0.08	0	0.042	−0.62	1.0	12.8
`quadrupole_bubble_001`	1.00	0.18	0	0.0095	−0.013	1.0	10.5
Schwarzschild puncture	0.29	0.00	0	—	−0.91	1.0	9.7

TABLE III. Candidates ranked by the gated objective S (Eq. 39). The three top gauge-angular search winners (Tier-1, $t = 2$, $N_{\rm full} = 64$) and the high-quality long-run validation of the best ($t = 16$, $N_{\rm full} = 192$) are compared to the reference seeds ($t = 16$, $N_{\rm full} = 64$). $\int \rho_- \, dV$ is the integrated required negative-energy density (geometric requirement from the vacuum Hamiltonian, Eq. 43; for the dipole winner it is a single localized core, Fig. 2); $\max c$ is the peak coordinate light speed on the initial slice; $F^{\rm ev}_{\rm op}$ is the evolved (persistence) operational-FTL benefit. The Tier-1 search S is evaluated over a short window and is therefore optimistic relative to the long-time value—the validated $t = 16$ row ($S = 16.75$) is the fair comparison with the seeds, and it still outscores Alcubierre. The non-spherical winners use *normal* matter ($\rho \geq 0$ constrained solve) with low residual exotic content; the reference wormhole/puncture seeds use phantom (exotic) initial data, and Alcubierre's matter sector is clean ($\int \rho_- \approx 0$) with its exoticity living in the geometry (effective EC).

Candidate	$t\,/\,N_{\rm full}$	$\int \rho_- \, dV$	$\max c$	$F^{\rm ev}_{\rm op}$	S
This work — gauge-angular non-spherical search (Tier-1)					
`eval_000188`	2 / 64	0.134	2.67	0.014	**18.30**
`eval_000183`	2 / 64	0.130	2.53	0.014	17.59
`eval_000170`	2 / 64	0.137	2.28	0.008	17.47
This work — long-run validation of the winner					
`eval_000188`	16 / 192	0.134	2.67 → 1.32	0.0056	16.75
Reference solutions ($t = 16$, constrained)					
Alcubierre $v = 0.5$	16 / 64	≈ 0	1.50	5×10^{-4}	14.9
Ellis–Bronnikov	16 / 64	phantom	1.00	0.042	12.8
`quadrupole_bubble_001`	16 / 64	phantom	1.00	0.0095	10.5
Schwarzschild puncture	16 / 64	phantom	1.00	—	9.7
Flat Minkowski	16 / 64	0	1.00	—	**0.0**

condition violation is required we now emit the geometric required energy density as a per-cell plot field,

$$\rho_{\rm req} = \frac{R + \frac{2}{3}K^2 - A_{ij}A^{ij}}{16\pi} = \frac{H_{\rm vac}}{16\pi}, \tag{43}$$

i.e. the Eulerian energy density that the Einstein constraints demand to source the *evolved* geometry, computed from the same vacuum Hamiltonian operator used for the constraint diagnostics (so the field integrates exactly to the tabulated $\int \rho_- \, dV$). Figure 2 maps it on the $z = 0$ slice of the validated winner. The result is unambiguous and somewhat surprising: the requirement is a *single, spatially localized exotic core* ($\rho_{\rm req} \approx -1.0 \times 10^{-3}$) sitting exactly on the dark compression crescent of χ, while the rest of the domain is essentially matter-free ($\rho_{\rm req} \approx 0$). The faint residual ring co-located with the outgoing K-wave is propagating gauge/constraint radiation, roughly 20–150× weaker than the core (its $\frac{2}{3}K^2$ piece is only $\sim 3\%$ of the local value; the rest is the R oscillation of the same outgoing pulse), and it leaves the grid rather than supporting the structure. Notably there is *no* comparable region of required ordinary matter: this is not a balanced positive/negative Alcubierre shell but a one-sided exotic lens.

b. How can it "move" with only an exotic component? It largely does *not* translate, and Fig. 2 explains why. The localized exotic core acts as a static spatial *lens*: its negative energy is what bends proper distance into the compress-ahead/expand-behind χ dipole, which is the geometric ingredient that opens a faster coordinate channel and produces the operational-FTL benefit (a signal crosses the lens in less coordinate time than the flat control). But a benefit on the initial slice is not propulsion. Translating that lens—carrying a passenger region along with it—requires a sustained *momentum* density j_i, since the off-diagonal Einstein equations

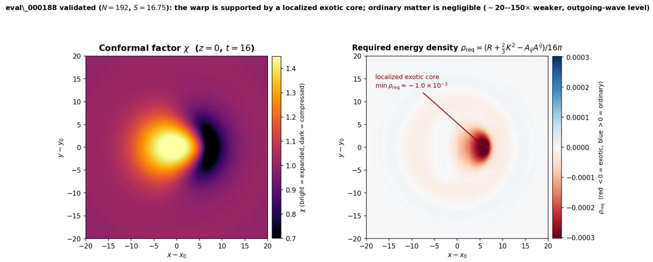

FIG. 2. **The warp dipole is held by a localized exotic core, with negligible ordinary matter.** Left: conformal factor χ at $z = 0$, $t = 16$ (bright = expanded, dark = compressed). Right: the geometric required energy density $\rho_{\rm req}$ of Eq. (43) on the same slice (red < 0 exotic, blue > 0 ordinary, shared symmetric scale). The energy-condition violation needed to sustain the geometry is concentrated in one compact core ($\min \rho_{\rm req} \approx -1.0 \times 10^{-3}$) coincident with the compression crescent; the surrounding domain is near-vacuum and the faint outer ring is the weak outgoing K-wave (junk radiation), not localized matter.

set $G_{0i} = 8\pi j_i$ and the momentum constraint relates j_i to the time-dependence of the shift (the warp "motor"). Our candidate has a near-clean momentum constraint ($\|M\|^{\rm fin}_{L^2} = 3.6 \times 10^{-5}$) and no driven shift, so there is no motor: the exotic core can hold the lens in place but cannot pump it across the grid. Consistently, the evolved channel coasts rather than propagates—peak coordinate speed decays ($2.67 \to 1.32$) and the structure stands and slowly relaxes rather than moving. In short, the exotic core is *necessary* to curve space into a shortcut, but *motion* additionally needs a co-moving momentum source; supplying that (a searchable translation shift / sustained j_i) is exactly the next degree of freedom identified in Sec. VIII.

Figure 3 shows the conformal factor χ for all four candidates in Table III that have frames. The conformal factor encodes the local stretching of space: the physical spatial metric is $\gamma_{ij} = \chi^{-1}\tilde{\gamma}_{ij}$, so regions with $\chi > 1$ (bright) have *expanded* proper distances—space is stretched—while regions with $\chi < 1$ (dark) have *compressed* proper distances—space is squeezed. For a flat, undeformed geometry $\chi = 1$ everywhere (uniform purple). The Alcubierre warp mechanism requires exactly this polarity: compressed space ahead of the passenger region and expanded space behind it.

VIII. DISCUSSION AND FUTURE WORK

This work turns exotic-metric study into a closed-loop GPU search over dynamical initial data. The current pipeline proposes constrained recipes, evolves them in `GRTeclyn`, scores stability/FTL/energy diagnostics, and validates promising candidates on longer runs. The first campaigns expose the central stability gap—optimizers can still favor geometries that merely outlive a short stop time—but also show that FTL scaffolding, non-triviality gating, growth metrics, evolved energy-condition checks, surrogate screening, and MAP-Elites can steer the search away from trivial flat space toward genuinely structured geometries like the dipole metric.

Several extensions would substantially expand the search capabilities.

a. Matter-first elliptic search. The largest recent improvement is making the constraint-satisfying path concrete: `GRTresna` now sits inside the optimizer loop and solves the Hamiltonian and momentum constraints for each proposed scalar-matter configuration before `GRTeclyn` evolves it. Per-lump exotic (phantom) matter and the maximal-slicing ($K = 0$) solve of Eq. (13) handle $\rho < 0$, and the multilevel safeguards of Sec. V G make the indefinite solve robust under full AMR for the mixed-matter configurations the search proposes. The current open work is no longer "make the solver run" but "validate and improve the candidates it discovers": replay the best evolved FTL precursor, extend its lifetime, test convergence, and add objectives that distinguish a stationary exotic lens from a co-moving transport bubble.

b. Topology search. The current parameterization cannot change topology: its non-spherical freedom lives in the gauge sector, where shift structure naturally favors warp-like channels. Wormholes and portals require a discrete controller that selects mouths, throats, modes, and boundary connectivity before the continuous optimizer refines each topology, echoing discrete/continuous decompositions in string-landscape search [3].

c. Additional degrees of freedom. Gauge-angular search produced a low-exotic directional channel. Next degrees of freedom include searchable radial placement of angular modes, full-z domains without reflective parity, persistence/co-motion rewards, soft barriers on near-trapped surfaces, and higher angular order with a dedicated translation shift.

d. Advanced parameterization. Future Level 3 replaces basis functions with an implicit neural decoder, optionally trained with Hamiltonian and momentum residuals as PINN losses [8]. A rule-based spacetime grammar can enforce asymptotic flatness, positivity, $K = {\rm const}$, $\Pi = 0$, and chosen topology before evaluation, reducing wasted GPU jobs in the spirit of grammar-constrained theory generation [1]. Level 4 is an LLM meta-agent that classifies failed runs, retunes exposed gauge/grid parameters, and proposes new modes while leaving the physics verdict to read-only diagnostics [2].

e. Multi-observer energy conditions. The $t = 0$ ANEC and tidal proxies are blind to pure-shift warps. Evaluating energy conditions from the evolved 4-metric via $T_{\mu\nu} = G_{\mu\nu}/8\pi$, e.g.

$$\Xi_W(\mathbf{x}) = \min_V T_{\mu\nu} V^\mu V^\nu, \qquad g_{\mu\nu} V^\mu V^\nu = -1, \tag{44}$$

recovers Alcubierre NEC/WEC violations missed by χ-sourced proxies. A Hawking–Ellis/Type-I hybrid margin following WARPAX [6] with fourth-order finite differences [7] feeds the score as $s^{\rm eff}_{\rm exotic}$.

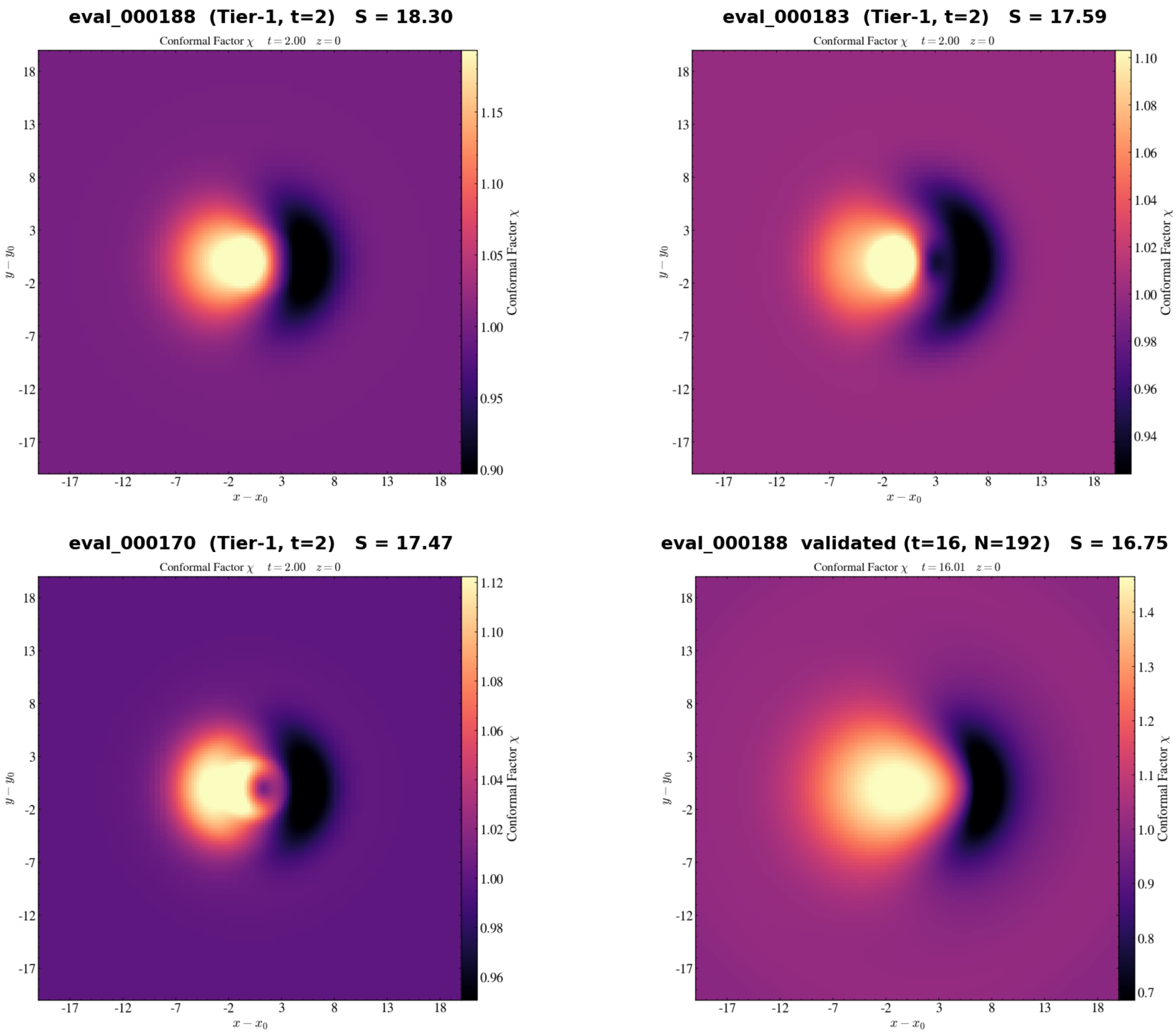

FIG. 3. **Directional χ channels of the top gauge-angular candidates** (conformal factor, $z = 0$ equatorial slice). The conformal factor χ encodes how much space is locally stretched ($\chi > 1$, bright/yellow = expanded) or compressed ($\chi < 1$, dark/black = squeezed); flat space has $\chi = 1$ (uniform purple). The three top-ranked Tier-1 search winners (`eval_000188`, `eval_000183`, `eval_000170` at $t = 2$, $N_{\text{full}} = 64$) and the high-quality long-run validation of the best (`eval_000188` at $t = 16$, $N_{\text{full}} = 192$) all share the same Alcubierre-like fore/aft dipole: a bright high-χ expansion lobe on the $-x$ side trailing a dark low-χ compression crescent on the $+x$ side—the "expand behind, compress ahead" polarity that a warp channel requires. The structure is reproducible across candidates and sharpens (rather than disperses) under long evolution at higher resolution. Each panel is annotated with its gated score S (cf. Table III).

f. Production-scale search. With surrogate screening, MAP-Elites, and Pareto extraction in place, production runs target a 1000-candidate constraint atlas plus a 50-generation CMA-ES campaign.

g. Analytical back-derivation. A numerical optimum is only a coefficient list. The final handoff reconstructs χ_{num}, fits a closed form by symbolic regression, solves the Hamiltonian constraint for $\phi(r)$, and reloads the analytic profile for verification. This can be automated as a multi-agent loop inspired by PhysAgent [4].

IX. WHAT IS STILL MISSING TO DISCOVER A NEW FTL METRIC

The pipeline above is, in our assessment, a working *engine* rather than a finished *discovery*. The candidates it has produced are promising precursors—a localized exotic lens with a weak evolved coordinate-superluminal channel (Sec. VII, Sec. V H)—but none yet qualifies as a new, validated, physically realizable faster-than-light geometry. This section consolidates, in priority order, the gaps that separate the present state from that goal. We group them into three classes: the objective rewards the wrong quantity, parts of the configuration space are unreachable, and discovered candidates cannot yet be trusted or extracted.

A. The objective rewards a static lens, not transport

The recurring empirical outcome is that every high-scoring geometry is a *standing* structure. The validated dipole winner is a static exotic lens that bends proper distance but does not translate (Sec. VII), and the matter-first winner `eval_000063` shows a shortcut without bulk motion. This is not a bug in the optimizer; it is the optimizer faithfully maximizing what we ask for. The operational measure $F_{\rm op}$ (Eq. (25)) rewards the *existence* of a faster coordinate channel on a slice, which a stationary lens already provides at minimal exotic cost. A genuine warp metric instead requires *transport*: a passenger region carried across a finite distance while the surrounding geometry remains quasi-stationary in its co-moving frame. Two coupled ingredients are missing.

a. A co-motion transport objective. We need a score term that tracks a co-moving worldtube and rewards a localized region traversing a baseline distance L in less proper (or coordinate) time than the flat control, *conditioned* on the local geometry being approximately stationary in the bubble frame. Without this conditioning the search will always prefer the cheaper static lens over a translating bubble.

b. A sustained *momentum driver—the "motor."* A translating bubble needs a sustained momentum density j_i, since $G_{0i} = 8\pi j_i$ ties the off-diagonal Einstein equations to the time-dependence of the shift. The matter-first `GRTresna` path already supplies searchable momentum-carrying matter and solves the momentum constraint for the associated frame-dragging shift (Sec. V); what is missing is *sustaining* it. Free evolution lets the seeded momentum disperse—our best candidate ends with a near-vacuum momentum constraint ($\|M\|^{\rm fin}_{L^2} = 3.6 \times 10^{-5}$) and no driven shift. A sustained momentum source or a translation Γ-driver target is the remaining degree of freedom that converts "a shortcut existed at $t = 0$" into "a bubble moved," and together with the transport objective above we regard it as the single highest-leverage missing element.

B. Unreachable regions of configuration space

a. Topology is frozen. All current freedom—gauge, conformal, and matter—lives on a fixed $\mathbb{R}^3$ slice topology. Traversable wormholes, portals, and handle/throat connectivities are therefore *not representable by any continuous parameter* the search exposes; an entire FTL class is excluded a priori. Reaching it requires the discrete topology controller of Sec. VIII that selects mouths, throats, and boundary connectivity before continuous refinement.

C. Trust, realism, and extraction

A high score is only a proxy; the search alone can never certify that a candidate is the sought solution. Confidence must come from survival under an increasingly demanding falsification ladder—constructed, non-trivial, operational, persistent, observer-robust, resolution-converged, and finally analytically reconstructed—of which the last stages are not yet automated.

a. Automated resolution convergence. Until $F_{\rm op}$, the constraint norms, and the exotic-energy integral are shown to be grid-independent, any candidate may be a discretization artifact—we have already observed near-degenerate metric cells masquerading as spectacular shortcuts. A standard multi-resolution auto-replay (rather than a manual check) is a prerequisite for any physical claim.

b. Quantum-inequality realism. The ANEC line proxy (Eq. (36)) penalizes the *amount* of negative energy but not its *physical attainability*. A geometry whose required exotic matter violates quantum-inequality (Ford–Roman) bounds is not a useful discovery; a quantum-inequality penalty would steer the search toward physically plausible exotic budgets.

c. Analytical back-derivation. As noted in Sec. VIII, a numerical optimum is a coefficient list, not a metric. Closing the loop with symbolic regression to a closed form, re-solving the constraints from that form, and reloading it for verification is what turns a high-scoring grid into a communicable, checkable spacetime.

In summary, the missing pieces are, in rough priority order: (i) a transport/co-motion objective coupled to a *sustained* momentum driver (`GRTresna` already supplies the momentum-carrying matter, but not a sustained driver); (ii) automated resolution-convergence validation; (iii) analytical back-derivation together with quantum-inequality realism; and (iv) a discrete topology controller to open the wormhole/portal class.

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